

Solution

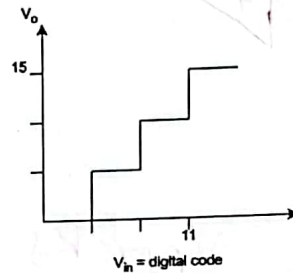
Assignment 3
Spring 25
CSE 350

Instruction:

- ✓ The solution must be in a hand-written format
- ✓ PDF file of the solution needs to be submitted through Google form

Question 1 of 2

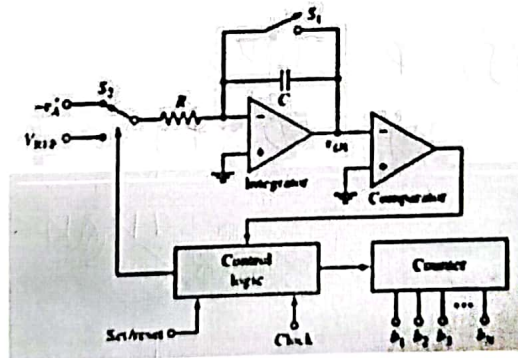
The figure below shows the input-output characteristics of a converter. The output of the converter is denoted as V_O and the input as V_{in} .



(a)	Identify the converter type and the number of bits used by the converter.
(b)	Design the converter. (Suppose, MSB resistance is R , assume $R_F = 2R$).

Question 2 of 2

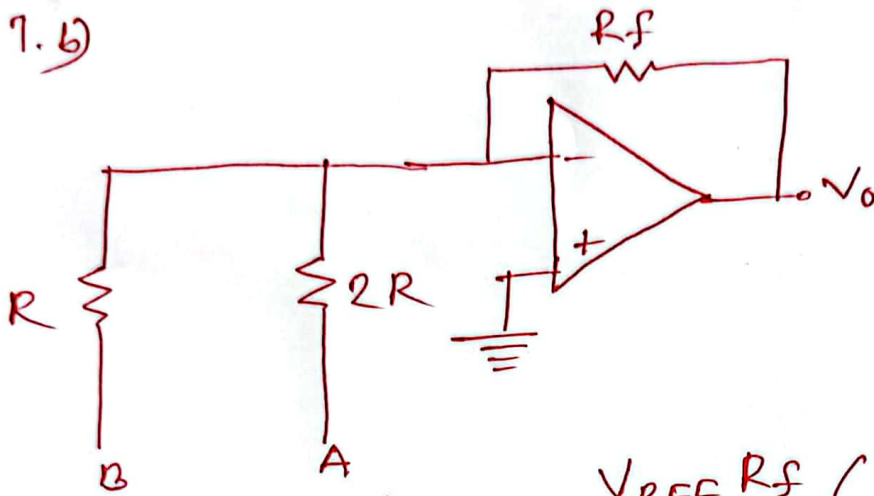
Consider the below Dual Slope ADC converter for answering the questions. The converter has $V_{REF} = 10\text{ V}$ and the output of the converter is 4 bits representation. A 1 MHz clock is used for the clock and counter circuits.



(a)	What will be the maximum reading time/ conversion time of the above Dual Slope A/D converter?
(b)	Find the maximum sampling frequency of the above circuit.
(c)	How can you make the sampling time 16 times without changing the output bit representation ?

1. a) DAC converter.
bits = 2

1. b)



$$V_o = - \frac{V_{REF} R_f}{R} \left(\frac{A}{2} + B \right)$$

From graph, 11, $V_o = 15$

$$15 = - \frac{V_{REF} \times 2R}{R} \left(\frac{1}{2} + 1 \right)$$

$$V_{REF} = -5V$$

if, $R = 10k$,

$$R_f = 2 \times R = 2 \times 10k = 20k$$

$$2. a) \quad T_c = \frac{1}{1M} = 1\mu s$$

$$n = 4$$

$$V_{REF} = 10V$$

$$T_{read max} = 2 \times 2^n \times T_c$$

$$= 2 \times 2^4 \times T_c$$

$$= 32\mu s$$

b)

$$F_{S max} = \frac{1}{2 \times 2^n \times T_c}$$

$$= \frac{1}{2 \times 2^4 \times 1\mu s}$$

$$= 31.25 kHz$$

c)

$$F_s = \frac{1}{2 \times 2^4 \times T_c}$$

$$F'_s = \frac{1}{2 \times 2^4 \times T'_c}$$

$$F'_s = 16 F_s$$

$$\frac{1}{2 \times 2^4 \times T'_c} = 16 \times \frac{1}{2 \times 2^4 \times T_c}$$

$$T_c = 16 T'_c$$

$$T'_c = \frac{1\mu s}{16}$$

$$\textcircled{C} \cdot F'_{clock} = \frac{1}{T'_c} = \frac{16}{1\mu s} = 16 MHz \quad (Ans)$$